Evaluation Form (1)

Title: Evaluation of the Proposed LLM-Based Agentic Architectural Framework for Smart Learning Environments.
Focus of the Interview: Learning Design Agent
Date:

We are conducting a short-term study to evaluate our proposed Large Language Model (LLM)-based agentic architectural framework that integrates three LLM-based agents (as shown in Figure 1) guided by SOLO (Structure of Observed Learning Outcome) taxonomy and Knowledge Components (KCs) into POLARIS which is a Smart Learning Environment (SLE). The framework is designed to strengthen the link between your learning design and learning analytics, capture the depth of student understanding through the SOLO taxonomy, and generate personalized, context-aware reactions for learners. This interview focuses on the *first LLM-based agent*, called the **Learning Design Agent**. Its purpose is to connect the teacher's learning design intentions with the analytics and reaction layers of the POLARIS. It makes sure that learning materials are turned into structured and pedagogically meaningful KCs. The agent analyzes teacher-provided materials, identifies the KCs, annotates them with mastery examples aligned to the SOLO taxonomy, defining the depth of reasoning expected, and map the KCs with learning activities.

Here is a short explanation of key concepts:

- ***LLMs****:* LLMs are advanced type of artificial intelligence trained on text data to understand and generate human-like language.
- ***KCs****:* These are small pieces of knowledge or skills that students are expected to learn. For example, in architecture, a KC could be "*Domestic origin of architecture according to Gottfried Semper*". A learning objective describes what students should achieve by the end of the course, such as explaining, analyzing, or designing something. In contrast, a KC represents a specific piece of knowledge or skill that students must acquire in order to successfully perform learning tasks. In simple terms, learning objectives define the goals of learning, while KCs are the knowledges or skills students need to master to reach those goals. They help connect teachers' intentions with how student learning is analyzed and tracked in learning systems.
- ***SOLO taxonomy****:* This is a framework to describe the depth of student understanding. It has five levels:
  1. Pre-structural – no real understanding.
  2. Uni-structural – focuses on just one detail.
  3. Multi-structural – knows several details but doesn't connect them.
  4. Relational – connects details into a meaningful whole.
  5. Extended Abstract – goes beyond, generalizes, or creates new ideas.

Activity ID	Title
A001	
A002	
A003	

Learning Design Agent

Step 1:

Please rate the 7 generated KCs in "Run A", their corresponding SOLO-aligned mastery examples and the suggested learning activity on a scale from 1 (Strongly Disagree) to 6 (Strongly Agree). If the Learning

Design Agent did not suggest the correct related learning activity for each KC, please refer to the table above and write the Activity ID/IDs that corresponds to each generated KC.

KC001: ...

Question	**Strongly Disagree (1)**	**Disagree (2)**	**Slightly Disagree (3)**	**Slightly Agree (4)**	**Agree (5)**	**Strongly Agree (6)**
The **KC001** proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						
The **KC001** proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for **KC001**.						
The suggested learning activity is appropriate for **KC001**.						
earning activity/activities corresponds to **KC001**? (A001 / A002 / A003)						

KC002: ...

Question	**Strongly Disagree (1)**	**Disagree (2)**	**Slightly Disagree (3)**	**Slightly Agree (4)**	**Agree (5)**	**Strongly Agree (6)**
The **KC002** proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						
The **KC002** proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for **KC002**.						
The suggested learning activity is appropriate for **KC002**.						
earning activity/activities corresponds to **KC002**? (A001 / A002 / A003)						

KC003: ...

Question	**Strongly Disagree (1)**	**Disagree (2)**	**Slightly Disagree (3)**	**Slightly Agree (4)**	**Agree (5)**	**Strongly Agree (6)**
The **KC003** proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						
The **KC003** proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for **KC003**.						
The suggested learning activity is appropriate for **KC003**.						
earning activity/activities corresponds to **KC003**? (A001 / A002 / A003)						

KC004: ...

Question	**Strongly Disagree (1)**	**Disagree (2)**	**Slightly Disagree (3)**	**Slightly Agree (4)**	**Agree (5)**	**Strongly Agree (6)**
The **KC004** proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						
The **KC004** proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for **KC004**.						
The suggested learning activity is appropriate for **KC004**.						
earning activity/activities corresponds to **KC004**? (A001 / A002 / A003)						

KC005: ...

Question	Strongly Disagree (1)	Disagree (2)	Slightly Disagree (3)	Slightly Agree (4)	Agree (5)	Strongly Agree (6)
The **KC005** proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						
The **KC005** proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for **KC005**.						
The suggested learning activity is appropriate for **KC005**.						
earning activity/activities corresponds to **KC005**? (A001 / A002 / A003)						

KC006: ...

Question	Strongly Disagree (1)	Disagree (2)	Slightly Disagree (3)	Slightly Agree (4)	Agree (5)	Strongly Agree (6)
The **KC006** proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						
The **KC006** proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for **KC006**.						
The suggested learning activity is appropriate for **KC006**.						
earning activity/activities corresponds to **KC006**? (A001 / A002 / A003)						

KC007: ...

Question	Strongly Disagree (1)	Disagree (2)	Slightly Disagree (3)	Slightly Agree (4)	Agree (5)	Strongly Agree (6)
The **KC007** proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						
The **KC007** proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for **KC007**.						
The suggested learning activity is appropriate for **KC007**.						
earning activity/activities corresponds to **KC007**? (A001 / A002 / A003)						

Step 2:

The following table present the KCs generated by three independent runs (Run A, Run B, Run C) of the Learning Design Agent using the same instructional materials.

In step 1, you have evaluated the KC in the first column of the table below, which belong to Run A. Now, please review the KCs generated in Runs B and C. Since each run produced its KCs independently, some may represent the same or very similar concepts, while others may differ slightly in wording or level of granularity. In the table below: Please identify and write the KC names from Runs B and C that correspond to the same or a very similar KC in Run A. After that, assign a score from 1 to 5 to indicate the level of overlap between them (1 = lowest overlap, 5 = highest overlap). Then, repeat the same steps starting from the KCs in Runs B and C. This task will help us evaluate the conceptual consistency of the generated KCs across the different runs.

Run A	Run B	Run C
Name: KC A1 Similar KC in Run B: ... Score (1-5): … Similar KC in Run C: ... Score (1-5): …	**Name: KC B1** Similar KC in Run A: ... Score (1-5): … Similar KC in Run C: ... Score (1-5): …	**Name: KC C1** Similar KC in Run A: ... Score (1-5): … Similar KC in Run B: ... Score (1-5): …
kc_id KC001	*kc_id* KC001	*kc_id* KC001
title Origen doméstico	*title*	*title*
description	*description*	*description*
target_SOLO_level	*target_SOLO_level*	*target_SOLO_level*
Related_learning_ activity	*Related_learning_ activity*	*Related_learning_ activity*
SOLO_level_mastery_ examples	*SOLO_level_mastery_ examples*	*SOLO_level_mastery_ examples*
media_context	*media_context*	*media_context*